The Mathematical Tools of Quantum Mechanics (Cohen–Tannoudji, Ch. II) Section F. Tensor products and applications

Beatriz Gonzalez del Rio

Outline

- A. One-particle wave function space
- B. State space and Dirac notation
- C. Representations in the state space
- D. Eigenvalue equations and observables
- E. Two key examples: $|\boldsymbol{r}\rangle$ and $|\boldsymbol{p}\rangle$; operators $\boldsymbol{R}$, $\boldsymbol{P}$
- F. Tensor products and applications

Why tensor products of state spaces?

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Question: How do we describe the state space of the *combined system*?

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To each pair of vectors

$$|\phi(1)\rangle \in \mathcal{E}_1, \qquad |\chi(2)\rangle \in \mathcal{E}_2,$$

we associate a vector

$$|\phi(1)\rangle \otimes |\chi(2)\rangle \in \mathcal{E}.$$

Basic properties of the tensor product

1) The tensor product is linear with respect to complex numbers:

$$\lambda \, |\phi(1)\rangle \otimes |\chi(2)\rangle = |\lambda\phi(1)\rangle \otimes |\chi(2)\rangle = |\phi(1)\rangle \otimes |\lambda\chi(2)\rangle \, .$$

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2) It is distributive with respect to vector addition:

$$|\phi(1)\rangle \otimes (|\chi_1(2)\rangle + |\chi_2(2)\rangle) = |\phi(1)\rangle \otimes |\chi_1(2)\rangle + |\phi(1)\rangle \otimes |\chi_2(2)\rangle\,.$$

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3) If $\{|u_i(1)\rangle\}$ and $\{|v_l(1)\rangle\}$ are basis of $\mathcal{E}_1$ and $\mathcal{E}_2$:

$$|u_i(1)\rangle \otimes |v_l(1)\rangle \text{ is a basis in } \mathcal{E}$$

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Key idea: The components of a product vector are *products* of components.

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Conclusion: Most states are *not* simple tensor products. However, the ones with most interest will be those that are.

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The scalar product in $\mathcal{E}$ of $|\phi(1)\chi(2)\rangle$ by $|\phi'(1)\chi'(2)\rangle$ is defined by:

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This definition is extended by linearity to all vectors of $\mathcal{E}$. If the bases of $\mathcal{E}_1$ and $\mathcal{E}_2$ are orthonormal, then:

$$\langle u_{i'}(1)v_{j'}(2)|u_i(1)v_j(2)\rangle = \delta_{ii'}\,\delta_{jj'}.$$

Tensor product of operators

Let $A(1)$ act in $\mathcal{E}_1$. Associated with it there is an operator $\tilde{A}(1)$ which acts in $\mathcal{E}$, called the extension of $A(1)$ in $\mathcal{E}$:

$$\tilde{A}(1)[|\phi(1)\rangle \otimes |\chi(2)\rangle] = [A(1)\,|\phi(1)\rangle] \otimes |\chi(2)\rangle$$

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This definition uniquely determines the operator on all of $\mathcal{E}$.

Projectors and commuting operators

The projector onto a tensor product vector, $|\phi(1)\chi(2)\rangle = |\phi(1)\rangle \otimes |\chi(2)\rangle$ is:

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Physical meaning: Observables acting on different subsystems are compatible.

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Important: Juxtaposition here *does not* mean an ordinary product.

Eigenvalue problems in tensor product spaces

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play a central role in the discussion of operators acting on composite systems.
In particular, we want to understand the eigenvalue problems of

$$A(1), \qquad B(2), \qquad C = A(1) + B(2)$$

when these operators act on the full space $\mathcal{E}$.

Eigenvalue equation of an extended operator

Assume that $A(1)$ is an observable in $\mathcal{E}_1$ with discrete spectrum:

$$A(1)\left|\phi_n^i(1)\right\rangle = a_n\left|\phi_n^i(1)\right\rangle; \qquad i = 1, 2, ..., g_n$$

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From the definition of the extension,

$$\tilde{A}(1)\left(\left|\phi_n^i(1)\right\rangle \otimes \left|\chi(2)\right\rangle\right) = a_n\left|\phi_n^i(1)\right\rangle \otimes \left|\chi(2)\right\rangle,$$

for any $\left|\chi(2)\right\rangle \in \mathcal{E}_2$.

Spectrum and degeneracy of $A(1)$

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However, each eigenvalue a_n becomes degenerate in $\mathcal{E}$.
An orthonormal basis of eigenvectors is given by

$$\left|\psi_{nl}^{i}\right\rangle = \left|\phi_{n}^{i}(1)\right\rangle \left|v_l(2)\right\rangle ,$$

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Thus, if $\dim \mathcal{E}_2 = N_2$, each eigenvalue a_n is $N_2 \times g_n$-fold degenerate.

Eigenvalue equation of $C = A(1) + B(2)$

Let $A(1)$ and $B(2)$ be observables with known eigenvalues:

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Since $A(1)$ and $B(2)$ commute, the tensor product vectors

$$|\phi_n(1)\rangle\,|\chi_p(2)\rangle$$

form a common eigenbasis.

Eigenvalues of $C = A(1) + B(2)$

Acting with C on a tensor product eigenvector:

$$C\big(\,|\phi_n(1)\rangle\,|\chi_p(2)\rangle\,\big) = (a_n + b_p)\,|\phi_n(1)\rangle\,|\chi_p(2)\rangle\,.$$

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Conclusion:

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If different pairs (n, p) give the same sum, degeneracy appears and not all eigenvectors are tensor products.

Complete sets of commuting observables

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forms a CSCO in the tensor product space $\mathcal{E}$.
Key idea: By combining complete commuting sets in $\mathcal{E}_1$ and $\mathcal{E}_2$, one obtains a complete set in $\mathcal{E}$.

From one-dimensional to three-dimensional state spaces

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associated with the observables Y and Z.
The three-dimensional state space is the tensor product

$$\mathcal{E}_{xyz} = \mathcal{E}_x \otimes \mathcal{E}_y \otimes \mathcal{E}_z.$$

Position basis and CSCO in three dimensions

A basis of $\mathcal{E}_{xyz}$ is obtained from tensor products:

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These kets are simultaneous eigenvectors of the extended operators:

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Therefore,

$$\{X, Y, Z\} \quad \text{is a C.S.C.O. in } \mathcal{E}_{xyz}.$$

Factorization and general states

Tensor product states of the form

$$|\phi\rangle\,|\chi\rangle\,|\omega\rangle$$

have factorized wave functions:

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Thus, $\psi(x,y,z)$ is a genuine function of three variables.

Separable Hamiltonians

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$$H_x \left|\phi_n\right\rangle = E_n^x \left|\phi_n\right\rangle, \quad H_y \left|\chi_p\right\rangle = E_p^y \left|\chi_p\right\rangle, \quad H_z \left|\omega_r\right\rangle = E_r^z \left|\omega_r\right\rangle.$$

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Then the eigenvalues and eigenvectors of H are

$$E_{npr} = E_n^x + E_p^y + E_r^z, \qquad \left|\psi_{npr}\right\rangle = \left|\phi_n\right\rangle \left|\chi_p\right\rangle \left|\omega_r\right\rangle.$$

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Conclusion: The state space of a two-particle system is the tensor product of the spaces which correspond to each of the particles.

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The square of the norm of $|\psi\rangle$ is:

$$\langle\psi|\psi\rangle = \int d^3r_1\, d^3r_2\, |\psi(\mathbf{r}_1, \mathbf{r}_2)|^2$$

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If $|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$, then

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \psi_1(\mathbf{r}_1)\, \psi_2(\mathbf{r}_2),$$

and the particles are *uncorrelated*.